

Supplementary Materials for “Interpretable and Efficient Penalized Multivariate Monomial Splines”

Matthew L. Davis

`matthewlouisdavis@gmail.com`

This supplement provides additional derivations, examples, simulation results, and theoretical results referenced in the main text. Sections are labeled S.A–S.G to distinguish them from the main-paper appendix. All equation, section, and definition numbering from the main paper is referenced with the prefix “main.”

S.A Multivariate Basis Example and Constraint Counting

The main paper (Section 2.1) illustrates the monomial basis in the single-predictor cubic case. Here we show how this generalizes to multiple predictors.

Consider $q = 3$ predictors $t, u, v \in \mathbb{R}$. For each observation i , a p -dimensional vector of up-to-quartic polynomial basis expansions with linear, quadratic, cubic, and quartic main effects, linear interactions, linear-quadratic interactions, and a single three-way linear interaction, can be written as follows. The same construction extends directly to more than

three predictors, although we restrict attention here to at most three-way interactions.

Let

$$\mathbf{x}_i = (1, t_i, t_i^2, t_i^3, t_i^4, u_i, u_i^2, u_i^3, u_i^4, v_i, v_i^2, v_i^3, v_i^4, t_i u_i, t_i v_i, u_i v_i, t_i u_i^2, u_i t_i^2, t_i v_i^2, v_i t_i^2, u_i v_i^2, v_i u_i^2, t_i u_i v_i)^\top.$$

Because the expansion in each predictor, conditional on the others, is in monomial form, taking first derivatives with respect to any predictor is straightforward. For example,

$$\frac{\partial}{\partial t} \mathbf{x}_i = (0, 1, 2t_i, 3t_i^2, 4t_i^3, 0, 0, 0, 0, 0, 0, 0, 0, u_i, v_i, 0, u_i^2, 2u_i t_i, v_i^2, 2v_i t_i, 0, 0, u_i v_i)^\top.$$

The three partial derivatives can be combined into a $q \times p$ matrix:

$$\mathbf{x}'_i = \mathbf{x}'_i(\mathbf{t}) = \begin{pmatrix} \frac{\partial}{\partial t} \mathbf{x}_i \\ \frac{\partial}{\partial u} \mathbf{x}_i \\ \frac{\partial}{\partial v} \mathbf{x}_i \end{pmatrix}.$$

Similarly,

$$\mathbf{x}''_i = \mathbf{x}''_i(\mathbf{t}) = \begin{pmatrix} \frac{\partial^2}{\partial t^2} \mathbf{x}_i + \frac{\partial^2}{\partial t \partial u} \mathbf{x}_i + \frac{\partial^2}{\partial t \partial v} \mathbf{x}_i \\ \frac{\partial^2}{\partial u^2} \mathbf{x}_i + \frac{\partial^2}{\partial t \partial u} \mathbf{x}_i + \frac{\partial^2}{\partial u \partial v} \mathbf{x}_i \\ \frac{\partial^2}{\partial v^2} \mathbf{x}_i + \frac{\partial^2}{\partial t \partial v} \mathbf{x}_i + \frac{\partial^2}{\partial u \partial v} \mathbf{x}_i \end{pmatrix}$$

is also $3 \times p$ matrix of predictor-specific “total” second derivatives, matching the convention used by `lgspline`: for each predictor, the pure second derivative and the mixed second derivatives involving that predictor are combined into one row. When more than one predictor is spline-expanded, the default construction then adds the corresponding first- and second-derivative rows elementwise,

$$\mathbf{d}_i = \mathbf{x}'_i(\mathbf{t}) + \mathbf{x}''_i(\mathbf{t}),$$

and uses $\mathbf{d}_i$ as the derivative matching block. This keeps the constraint system smaller than when constraints are imposed separately. Since the first derivative always sets the linear term of $\mathbf{x}'$ to 1, regardless of knot location, and to 0 for $\mathbf{x}''(\mathbf{t})$, the two vectors are linearly independent and it is never the case that $\mathbf{x}'(\mathbf{t}) = -\mathbf{x}''(\mathbf{t})$, thus $\mathbf{d}_i \neq \mathbf{0}$ and some degenerate constraints are avoided. The separate construction remains as a non-default option in `lgspline`.

The constraints imply that at each knot separating partitions $\mathcal{P}_k$ and $\mathcal{P}_{k^*}$ with expansions evaluated at the shared boundary point:

$$\mathbf{x}_{k,k^*}^\top (\boldsymbol{\beta}_k - \boldsymbol{\beta}_{k^*}) = \mathbf{d}_{k,k^*}^\top (\boldsymbol{\beta}_k - \boldsymbol{\beta}_{k^*}) = 0.$$

The constraints can be encoded in a matrix $\mathbf{A} \in \mathbb{R}^{P \times R}$ such that the smoothing constraints are met when $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$. Additional linear constraints can also be included, or specific smoothness constraints removed, by adding or deleting columns from the constraint matrix. For illustration, let $K = 3$ knots define 4 partitions separated by three knot locations $\mathbf{t}_{1,2}, \mathbf{t}_{2,3}, \mathbf{t}_{3,4}$ with expansions $\mathbf{x}_{1,2} = \mathbf{x}(\mathbf{t}_{1,2}), \mathbf{x}_{2,3} = \mathbf{x}(\mathbf{t}_{2,3}), \mathbf{x}_{3,4} = \mathbf{x}(\mathbf{t}_{3,4})$:

$$\mathbf{A}^\top \boldsymbol{\beta} = \begin{pmatrix} \mathbf{x}_{1,2}^\top & -\mathbf{x}_{1,2}^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{d}_{1,2}^\top & -\mathbf{d}_{1,2}^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{x}_{2,3}^\top & -\mathbf{x}_{2,3}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{2,3}^\top & -\mathbf{d}_{2,3}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{x}_{3,4}^\top & -\mathbf{x}_{3,4}^\top \\ \mathbf{0} & \mathbf{0} & \mathbf{d}_{3,4}^\top & -\mathbf{d}_{3,4}^\top \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}.$$

S.A.2 Constraint counting: $R < P$

The main paper states that $\mathbf{A}$ is typically underdetermined in applications. This follows from a simple count. Let ϕ denote the average number of knots touching a partition. Under

the default combined derivative construction used when more than one predictor is spline-expanded, each knot contributes one fitted-value constraint and q derivative constraints, giving

$$R = \phi K(q + 1),$$

where K is the number of knots. If first- and second-derivative constraints are imposed separately, the same count becomes $R_{\text{sep}} = \phi K(2q + 1)$.

The coefficient dimension is

$$P = (K + 1)p, \quad p \geq 4q + 1,$$

where $4q + 1$ counts an intercept plus linear, quadratic, cubic, and quartic terms for each spline-expanded predictor. Interactions only increase p , but do not create additional knots or derivative matching conditions. Therefore they make the inequality $R < P$ easier to satisfy.

Using the lower bound $p \geq 4q + 1$, the default combined-derivative system is underdetermined whenever

$$\phi K(q + 1) < (K + 1)(4q + 1) \quad \Longleftrightarrow \quad \phi < \left(\frac{K + 1}{K} \right) \left(\frac{4q + 1}{q + 1} \right).$$

Since $(K + 1)/K > 1$, a simple sufficient condition is

$$\phi \leq \frac{4q + 1}{q + 1},$$

which removes the dependence on K . For a single predictor, $\phi = 1$ because each knot separates two adjacent intervals. Even under the more conservative separate derivative

construction,

$$\phi \leq \frac{4q+1}{2q+1} = \frac{5}{3} \quad (q=1),$$

so the usual single-predictor spline has $R < P$ once quadratic, cubic, or quartic terms are included.

The only caveat is geometric. In higher dimensions, a partition can have many neighbors, so ϕ can grow with q and the bound can fail. This is why `lgspline` combines derivative rows by default for multivariate splines and allows selected derivative constraints to be removed. Retaining only fitted-value continuity gives an even smaller constraint system, which can be useful when many predictors or many neighboring partitions are present.

S.B Woodbury Factorization for Structured Correlation

This supplement derives the structured-correlation acceleration described in the main paper (Section 3.3). The goal is to preserve the transformed-OLS projection while avoiding a dense $P \times P$ factorization while preserving as many opportunities for parallelism as possible.

Write

$$\mathbf{G}^{-1} = \mathbf{X}^T \mathbf{V}^{-1} \mathbf{X} + \mathbf{\Lambda} = \mathbf{G}_{\text{on}}^{-1} + \mathbf{G}_{\text{off}}^{-1}, \quad (1)$$

where $\mathbf{G}_{\text{on}}^{-1}$ contains the $K+1$ block-diagonal $p \times p$ pieces and $\mathbf{G}_{\text{off}}^{-1}$ contains the remaining cross-partition terms. The superscript -1 is part of this additive inverse-scale notation; $\mathbf{G}_{\text{off}}^{-1}$ is not the inverse of a separately defined $\mathbf{G}_{\text{off}}$. Without structured correlation, $\mathbf{G}_{\text{off}}^{-1} = \mathbf{0}$. The “Woodbury factorization” approach is useful when $\mathbf{G}_{\text{off}}^{-1}$ is sparse and low rank relative to the full coefficient dimension, which is common when many partitions are present but each correlated subject trajectory spans only a few of them. Woodbury’s identity is

then used to exploit this structure.

Let $\mathbf{E}\mathbf{J}\mathbf{E}^\top$ denote a reduced spectral factorization of $\mathbf{G}_{\text{off}}^{-1}$ retaining only the s nonzero eigen-directions, with $\mathbf{J}$ diagonal and entries ± 1 . The Woodbury identity gives

$$\mathbf{G} = \mathbf{G}_{\text{on}} - \mathbf{G}_{\text{on}}\mathbf{E}(\mathbf{J}^{-1} + \mathbf{E}^\top\mathbf{G}_{\text{on}}\mathbf{E})^{-1}\mathbf{E}^\top\mathbf{G}_{\text{on}} \quad (2)$$

$$= \mathbf{G}_{\text{on}}^{1/2} \underbrace{\left[\mathbf{I} - \mathbf{G}_{\text{on}}^{1/2}\mathbf{E}(\mathbf{J}^{-1} + \mathbf{E}^\top\mathbf{G}_{\text{on}}\mathbf{E})^{-1}\mathbf{E}^\top\mathbf{G}_{\text{on}}^{1/2} \right]}_{\mathbf{F}} \mathbf{G}_{\text{on}}^{1/2}. \quad (3)$$

Define $\mathbf{N} = \mathbf{G}_{\text{on}}^{1/2}\mathbf{E}$. Then

$$\mathbf{F} = \mathbf{I} - \mathbf{N}(\mathbf{J}^{-1} + \mathbf{N}^\top\mathbf{N})^{-1}\mathbf{N}^\top. \quad (4)$$

Applying Woodbury once more yields

$$\mathbf{F} = (\mathbf{I} + \mathbf{N}\mathbf{J}\mathbf{N}^\top)^{-1}. \quad (5)$$

Substituting the QR decomposition $\mathbf{N} = \mathbf{Q}\mathbf{R}$, with $\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top$, gives the inverse of $\mathbf{F}$ as:

$$\mathbf{F}^{-1} = \mathbf{I} + \mathbf{N}\mathbf{J}\mathbf{N}^\top = (\mathbf{I} - \mathbf{P}) + \mathbf{P} + \mathbf{Q}\mathbf{R}\mathbf{J}\mathbf{R}^\top\mathbf{Q}^\top = (\mathbf{I} - \mathbf{P}) + \mathbf{Q} \underbrace{(\mathbf{I}_s + \mathbf{R}\mathbf{J}\mathbf{R}^\top)}_{\mathbf{S}} \mathbf{Q}^\top. \quad (6)$$

The matrices $\mathbf{I} - \mathbf{P}$ and $\mathbf{Q}\mathbf{S}\mathbf{Q}^\top$ belong to orthogonal subspaces so $(\mathbf{I} - \mathbf{P})\mathbf{Q}\mathbf{S}\mathbf{Q}^\top = \mathbf{0}$, $\mathbf{Q}$ is orthonormal so $\mathbf{Q}\mathbf{S}^{1/2}\mathbf{Q}^\top\mathbf{Q}\mathbf{S}^{1/2}\mathbf{Q}^\top = \mathbf{Q}\mathbf{S}\mathbf{Q}^\top$, and $\mathbf{I} - \mathbf{P}$ is idempotent so $(\mathbf{I} - \mathbf{P})(\mathbf{I} - \mathbf{P}) = \mathbf{I} - \mathbf{P}$, and the matrix square roots follow conveniently:

$$\mathbf{F}^{-1/2} = (\mathbf{I} - \mathbf{P}) + \mathbf{Q}\mathbf{S}^{1/2}\mathbf{Q}^\top, \quad \mathbf{F}^{1/2} = (\mathbf{I} - \mathbf{P}) + \mathbf{Q}\mathbf{S}^{-1/2}\mathbf{Q}^\top.$$

The full transformed-OLS equivalence follows by substituting the Woodbury factoriza-

tion directly into the projection. Define

$$\mathbf{F}^{1/2} = (\mathbf{I} - \mathbf{P}) + \mathbf{Q}\mathbf{S}^{-1/2}\mathbf{Q}^\top, \quad \mathbf{L} = \mathbf{G}_{\text{on}}^{1/2}\mathbf{F}^{1/2}, \quad \tilde{\mathbf{U}} = \mathbf{I} - \mathbf{G}\mathbf{A}(\mathbf{A}^\top\mathbf{G}\mathbf{A})^{-1}\mathbf{A}^\top.$$

Since

$$\mathbf{G}^{-1} = \mathbf{G}_{\text{on}}^{-1} + \mathbf{G}_{\text{off}}^{-1}, \quad \mathbf{G}_{\text{off}}^{-1} = \mathbf{E}\mathbf{J}\mathbf{E}^\top, \quad \mathbf{F}^{-1} = \mathbf{I} + \mathbf{G}_{\text{on}}^{1/2}\mathbf{E}\mathbf{J}\mathbf{E}^\top\mathbf{G}_{\text{on}}^{1/2} = \mathbf{I} - \mathbf{P} + \mathbf{Q}\mathbf{S}\mathbf{Q}^\top,$$

we have $\mathbf{L}\mathbf{L}^\top = \mathbf{G}_{\text{on}}^{1/2}\mathbf{F}\mathbf{G}_{\text{on}}^{1/2} = \mathbf{G}$. Thus, for $\mathbf{w} = \mathbf{X}^\top\mathbf{V}^{-1}\mathbf{y}$ and $\hat{\boldsymbol{\beta}} = \mathbf{G}\mathbf{w}$,

$$\begin{aligned} \tilde{\mathbf{U}}\hat{\boldsymbol{\beta}} &= [\mathbf{I} - \mathbf{G}\mathbf{A}(\mathbf{A}^\top\mathbf{G}\mathbf{A})^{-1}\mathbf{A}^\top]\hat{\boldsymbol{\beta}} \\ &= \mathbf{G}_{\text{on}}^{1/2}\mathbf{F}^{1/2}\left[\mathbf{F}^{1/2}\mathbf{G}_{\text{on}}^{1/2}\mathbf{w} - \mathbf{F}^{1/2}\mathbf{G}_{\text{on}}^{1/2}\mathbf{A}\{\mathbf{A}^\top\mathbf{G}_{\text{on}}^{1/2}\mathbf{F}\mathbf{G}_{\text{on}}^{1/2}\mathbf{A}\}^{-1}\mathbf{A}^\top\mathbf{G}_{\text{on}}^{1/2}\mathbf{F}\mathbf{G}_{\text{on}}^{1/2}\mathbf{w}\right] \\ &= \mathbf{G}_{\text{on}}^{1/2}[(\mathbf{I} - \mathbf{P}) + \mathbf{Q}\mathbf{S}^{-1/2}\mathbf{Q}^\top]\left[\mathbf{y}^* - \mathbf{X}^*(\mathbf{X}^{*\top}\mathbf{X}^*)^{-1}\mathbf{X}^{*\top}\mathbf{y}^*\right] \\ &= \mathbf{G}_{\text{on}}^{1/2}[(\mathbf{I} - \mathbf{P}) + \mathbf{Q}\mathbf{S}^{-1/2}\mathbf{Q}^\top]\mathbf{r}^* = \mathbf{L}\mathbf{r}^* = \tilde{\boldsymbol{\beta}}, \end{aligned} \tag{7}$$

where

$$\mathbf{y}^* = [(\mathbf{I} - \mathbf{P}) + \mathbf{Q}\mathbf{S}^{-1/2}\mathbf{Q}^\top]\mathbf{G}_{\text{on}}^{1/2}\mathbf{w}, \quad \mathbf{X}^* = [(\mathbf{I} - \mathbf{P}) + \mathbf{Q}\mathbf{S}^{-1/2}\mathbf{Q}^\top]\mathbf{G}_{\text{on}}^{1/2}\mathbf{A}.$$

This is the same transformed-OLS argument as in Appendix A of the main text, with the block-diagonal term and the rank- s Woodbury correction carried through explicitly.

Apart from the low-rank orthonormalization of $\mathbf{N}$, the projection step only requires a QR decomposition of the $P \times R$ matrix $\mathbf{X}^*$. If the constraints are nonzero, $\mathbf{A}^\top\boldsymbol{\beta} = \mathbf{c}$, the same particular-solution adjustment used in the main paper (Section 2.2) is added after projection.

S.C REML for Correlation Parameters

When a working correlation $\mathbf{V} = \mathbf{V}(\boldsymbol{\rho})$ is present, $\boldsymbol{\rho}$ is estimated by minimizing a negative restricted maximum likelihood (REML) criterion scaled by $1/N$:

$$\begin{aligned} -\log \mathcal{L}_R(\boldsymbol{\rho}, \sigma^2) = & \frac{1}{N} \left[-\log |\mathbf{V}^{-1/2}| + \frac{N}{2} \log \sigma^2 \right. \\ & + \frac{1}{2\sigma^2} \left\| \mathbf{V}^{-1/2} \mathbf{D}^{1/2} \tilde{\mathbf{W}}^{-1/2} (\mathbf{y} - \tilde{\boldsymbol{\mu}}) \right\|^2 \\ & \left. + \frac{1}{2} \log |(\sigma^2 \tilde{\mathbf{U}} \tilde{\mathbf{G}} \tilde{\mathbf{U}}^\top)^{-1}|^+ \right], \end{aligned}$$

where $|\cdot|^+$ is the generalized determinant, the product of the nonzero eigenvalues, $\tilde{\boldsymbol{\mu}} =$

$g^{-1}(\mathbf{X}\tilde{\boldsymbol{\beta}}) = \mathbb{E}[\mathbf{y}]$ for link function g , $\tilde{\mathbf{W}} = \begin{pmatrix} \ddots & & \\ & [V(\tilde{\mu}_i)\{g'(\tilde{\mu}_i)\}^2]^{-1} & \\ & & \ddots \end{pmatrix}$ is a diagonal

matrix of GLM weights, $\mathbf{D}$ is a diagonal matrix of user-supplied custom weights, and $\tilde{\mathbf{G}}$ is the inverse penalized information matrix from the whitened, weighted system. This reduces to classical unbiased REML estimation for Gaussian responses with identity link and no penalization or constraint. The dispersion estimate used by the implementation is $\tilde{\sigma}^2 = \frac{\|\mathbf{V}^{-1/2} \mathbf{D}^{1/2} \tilde{\mathbf{W}}^{-1/2} (\mathbf{y} - \tilde{\boldsymbol{\mu}})\|^2}{N - \text{tr}(\mathbf{H})}$ which reduces to the usual REML estimate for Gaussian response and identity link, and reduces to a re-scaled method-of-moment estimator for non-Gaussian response. This approximation differs from a typical generalization of REML to non-Gaussian response. It is convenient for implementation in `lgspline` because it only requires the moments of the distribution to be known without specifying the full log-likelihood (or quasi-likelihood etc.), and gradients with respect to $\boldsymbol{\rho}$ are mostly preserved for gradient-based optimization.

S.D GLM Extension: Consistency and Asymptotic Normality

This section gives the large-sample motivation for the GLM extension. The actual constrained penalized MLE is not derived directly, so the argument follows the constrained-projection logic from the Gaussian case after replacing the Gaussian precision by the fitted GLM information matrix.

Under standard regularity conditions for a correctly specified penalized GLM, let $\hat{\beta}$ denote the unconstrained estimator. Then

$$\hat{\beta} \xrightarrow{p} \beta^*, \quad (8)$$

$$\mathbf{G}^{-1/2}(\hat{\beta} - \beta^*) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma), \quad (9)$$

for a target β^* satisfying $\mathbf{A}^\top \beta^* = \mathbf{0}$ (so $\mathbf{U}(\beta)\beta^* = \beta^*$ for any β), and a positive-definite covariance matrix Σ . The one-step constrained estimator is

$$\tilde{\beta}^{(1)} = \mathbf{U}(\hat{\beta})\hat{\beta}, \quad (10)$$

where $\mathbf{U}(\cdot)$ is the projection matrix built from the fitted information matrix. If $\mathbf{U}(\hat{\beta})$ converges in probability to $\mathbf{U}^* = \mathbf{U}(\beta^*)$, the continuous mapping theorem gives

$$\mathbf{U}(\hat{\beta})\hat{\beta} \xrightarrow{p} \mathbf{U}^*\beta^* = \beta^*. \quad (11)$$

This gives consistency of the one-step constrained estimator. Under the same assumptions, Slutsky's theorem gives asymptotic normality for any fixed linear combination:

$$\sqrt{N} \{ \mathbf{x}^\top (\tilde{\beta}^{(1)} - \beta^*) \} \xrightarrow{d} \mathcal{N}(0, \tau^2), \quad (12)$$

for some $\tau^2 > 0$.

The estimator returned by `lgspline` recomputes $\mathbf{U}$ at the most recent constrained estimate. The same argument applies by induction. Since $\tilde{\boldsymbol{\beta}}^{(1)} \rightarrow \boldsymbol{\beta}^*$,

$$\tilde{\boldsymbol{\beta}}^{(2)} = \mathbf{U}(\tilde{\boldsymbol{\beta}}^{(1)})\hat{\boldsymbol{\beta}} \rightarrow \mathbf{U}^*\boldsymbol{\beta}^* = \boldsymbol{\beta}^* \quad (13)$$

and the argument repeats for later iterations. The same Slutsky argument gives asymptotic normality of fixed linear combinations, with the same limiting distribution as the one-step estimator.

S.E Simulation DGPs and Surface Visualizations

This section gives the exact DGPs used in the simulation study. For $q = 1$, the target is the Donoho-Johnstone Doppler function (Donoho and Johnstone, 1994) on $[0, 1]$, scaled by 5:

$$f_1(t) = 5\sqrt{t(1-t)} \sin(2\pi \cdot 1.05 / (t + 0.05)). \quad (14)$$

For $q = 2$, the observed predictors t_1 and t_2 are generated on unequal scales. Let

$$u_1 = (t_1/36)^{1/1.65}, \quad u_2 = ((t_2 + 3)/12)^{1/0.62}. \quad (15)$$

The quantity u_1 is drawn from $\text{Beta}(1.4, 2.4)$, except with probability 0.25 it is drawn from $\text{Beta}(4.5, 1.8)$ instead. The quantity u_2 is drawn from $\text{Beta}(0.95, 2.2)$, except with probability 0.22 it is drawn from $\text{Beta}(5.5, 2.0)$ instead. This generates skewed observed

predictors on $t_1 \in [0, 36]$ and $t_2 \in [-3, 9]$. The true surface is

$$\begin{aligned}
f_2(t_1, t_2) = & 0.75 \sin(2\pi u_1) \cos(1.5\pi u_2) \\
& + 1.30 \exp\{-55[u_2 - 0.58 - 0.16 \sin(2.4\pi u_1)]^2\} \\
& + 1.35 \exp\{-38[(u_1 - 0.23)^2 + 0.50(u_2 - 0.78)^2]\} \\
& - 1.20 \exp\{-70[(u_1 - 0.72)^2 + (u_2 - 0.28)^2]\} \\
& + 0.75(u_1 - 0.5)(u_2 - 0.5) + 0.22|u_1 - 0.62| \\
& - 0.22(u_2 - 0.72)_+ + 0.28(u_1 + 0.55u_2 - 1.08)_+,
\end{aligned}$$

where $(a)_+ = \max(a, 0)$. The surface includes a broad oscillation, a curved ridge, two localized features, and light nonsmooth terms.

For $q = 4$, independent standard normal variables are first generated, and dependence is imposed through the recursion

$$\begin{aligned}
r_1 &= e_1, \\
r_2 &= 0.55 r_1 + \sqrt{1 - 0.55^2} e_2, \\
r_3 &= -0.35 r_1 + 0.45 r_2 + \sqrt{1 - 0.35^2} e_3, \\
r_4 &= 0.40 r_1 - 0.25 r_3 + \sqrt{1 - 0.40^2} e_4,
\end{aligned}$$

where $e_1, \dots, e_4$ are independent standard normal variables. The normal-CDF transforms $\Phi(r_j)$ are then used to construct the observed predictors

$$\begin{aligned}
t_1 &= 55 \Phi(r_1)^{1.55}, \quad t_2 = -4 + 16 \Phi(r_2)^{0.70}, \\
t_3 &= 7 \Phi(r_3)^{1.25}, \quad t_4 = -2.2 + 7 \Phi(r_4)^{0.82} + 0.35 \{\Phi(r_4) - 0.58\}_+^2.
\end{aligned}$$

Before this final transformation, $\Phi(r_2)$ is replaced by an independent Beta(1.0, 3.2) draw with probability 0.20, and $\Phi(r_4)$ is replaced by an independent Beta(6.0, 2.2) draw with

probability 0.22. The inverse transforms used to write the regression surface are

$$\begin{aligned} u_1 = s_1(t_1) &= (t_1/55)^{1/1.55}, & u_2 = s_2(t_2) &= ((t_2 + 4)/16)^{1/0.70}, \\ u_3 = s_3(t_3) &= (t_3/7)^{1/1.25}, & u_4 = s_4(t_4), \end{aligned} \tag{16}$$

where $s_4(t_4)$ is the unique value in $[0, 1]$ satisfying $t_4 = -2.2 + 7 u_4^{0.82} + 0.35 (u_4 - 0.58)_+^2$, and let $z_j = u_j - 0.5$. The true surface is

$$\begin{aligned} f_4(t_1, t_2, t_3, t_4) = & 0.65 \sin(2\pi u_1) + 0.55 \cos(2\pi u_2) + 0.45 \sin(2\pi(u_3 + 0.25 u_4)) \\ & + 0.65 z_2^2 - 0.45 z_4^2 + 1.15 z_1 z_2 - 0.95 z_2 z_3 + 1.70 z_1 z_3 z_4 \\ & + 1.25 \exp\{-55[(u_1 - 0.75)^2 + (u_4 - 0.25)^2]\} \\ & - 1.10 \exp\{-45[(u_2 - 0.25)^2 + (u_3 - 0.70)^2]\} \\ & + 0.18 |u_3 - 0.55| + 0.28 (u_1 - 0.72)_+ (0.35 - u_2)_+ \\ & + 0.16 (u_3 + u_4 - 1.15)_+. \end{aligned}$$

S.F Fixed-Penalty Simulation Results

This supplement reports RMSE for the simulation study in the main text when data-adaptive smoothing selection is skipped. The `mgcv` and truncated-polynomial fits use fixed smoothing parameters $\lambda = 10^{-4}$, while `lgspline` uses its default penalty values. All other settings (sample sizes, noise levels, replication count $M = 24$, number of cores) are identical to the tuned case.

The qualitative patterns are similar to the tuned results, with the fixed-penalty regime isolating per-fit cost more directly. `lgspline` is most accurate in all $q = 1$ cells, most $q = 2$ cells, and the largest low-noise $q = 4$ cell. In that largest low-noise $q = 4$ cell, fixed-penalty `lgspline` has RMSE 0.107 compared with 0.108 under automatic tuning, indicating that

tuning is not guaranteed to improve every finite-sample average. The truncated polynomial is fastest, as in the main text, but that comparison should be read in light of its simpler dense implementation and tuning setup.

Table 1: RMSE against f_{true} (mean (SD)) under the fixed-penalty regime over 24 replications. Lowest feasible RMSE in each column is **bold**.

σ	Method	$q = 1$			$q = 2$			$q = 4$		
		$N=500$	$N=5k$	$N=50k$	$N=500$	$N=5k$	$N=50k$	$N=500$	$N=5k$	$N=50k$
0.5	lgspline	0.678 (0.061)	0.672 (0.047)	0.626 (0.022)	0.231 (0.014)	0.121 (0.005)	0.102 (0.004)	0.396 (0.012)	0.162 (0.004)	0.107 (0.002)
	Cubic RS	0.713 (0.068)	0.703 (0.027)	0.692 (0.013)	0.335 (0.012)	0.319 (0.004)	0.319 (0.002)	0.258 (0.012)	0.140 (0.003)	0.122 (0.001)
	TPRS	0.713 (0.044)	0.724 (0.010)	0.716 (0.003)	0.242 (0.016)	0.212 (0.005)	0.211 (0.003)	0.267 (0.010)	0.183 (0.003)	0.174 (0.003)
	Trunc. poly	0.825 (0.042)	0.752 (0.014)	0.699 (0.010)	0.424 (0.013)	0.413 (0.005)	0.413 (0.002)	0.236 (0.012)	0.189 (0.003)	0.182 (0.001)
2.0	lgspline	0.761 (0.053)	0.681 (0.045)	0.618 (0.018)	0.842 (0.056)	0.299 (0.018)	0.134 (0.005)	1.552 (0.080)	0.538 (0.019)	0.191 (0.005)
	Cubic RS	0.794 (0.059)	0.710 (0.038)	0.698 (0.014)	0.550 (0.045)	0.351 (0.009)	0.321 (0.002)	0.959 (0.068)	0.337 (0.021)	0.156 (0.004)
	TPRS	0.788 (0.037)	0.729 (0.012)	0.717 (0.003)	0.511 (0.041)	0.259 (0.008)	0.215 (0.004)	0.903 (0.053)	0.333 (0.018)	0.195 (0.004)
	Trunc. poly	0.881 (0.034)	0.755 (0.021)	0.703 (0.012)	0.594 (0.036)	0.435 (0.009)	0.416 (0.001)	0.658 (0.071)	0.279 (0.018)	0.193 (0.002)

S.G Single-Predictor Base R Reference Implementation

This supplement references a compact base R reference implementation of the constrained monomial spline procedure for a single predictor with a cubic basis over three partitions. The script uses only base R together with the helper functions `matsqrt` and `matinvsqrt` from `lgspline` for symmetric matrix square roots, and it follows the four-step transformed-OLS projection described in the main text. An extension to logistic regression is also included. Extensions to multiple predictors, interactions, other GLM responses, and structured correlation follow the same template but are more verbose; the full implementation is in the `lgspline` package. The corresponding script `simple_illustration.R` is distributed with the supplementary materials.

Figure 2 shows the unconstrained and constrained fits produced by this script on a noisy test signal with $N = 150$ points. The unconstrained fits are independent cubic polynomials on each of the three partitions with visible jumps and kinks at the knots $t = \pm 2$; the constrained fit is everywhere twice continuously differentiable while retaining the local

cubic-polynomial representation within each partition.

References

Donoho, D. L. and Johnstone, I. M. (1994). Ideal spatial adaptation by wavelet shrinkage.
Biometrika, 81(3):425–455.

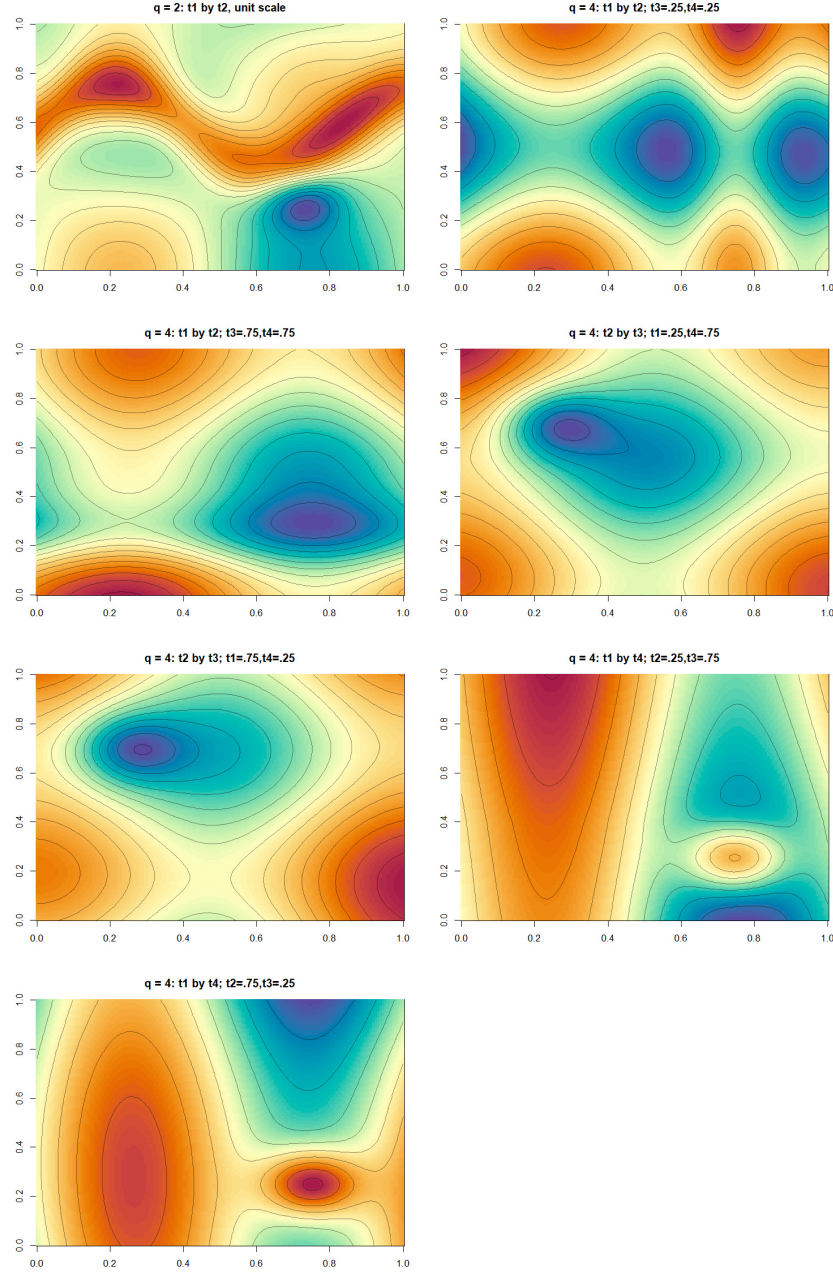


Figure 1: Surface slices of the multivariate DGPs used in the simulation study. The first panel shows the full $q = 2$ surface on the unit-scale predictor coordinates. The remaining panels show two-dimensional slices of the $q = 4$ surface with the other predictors fixed at the displayed values. Panel titles list the two unit-scale predictors shown on the horizontal and vertical axes and, for $q = 4$, the fixed unit-scale values of the remaining predictors. Topographic color shading encodes relative true surface height within each panel, and black contour lines show equal-height curves. Observed predictors are generated from these unit-scale coordinates by the monotone range transformations described above.

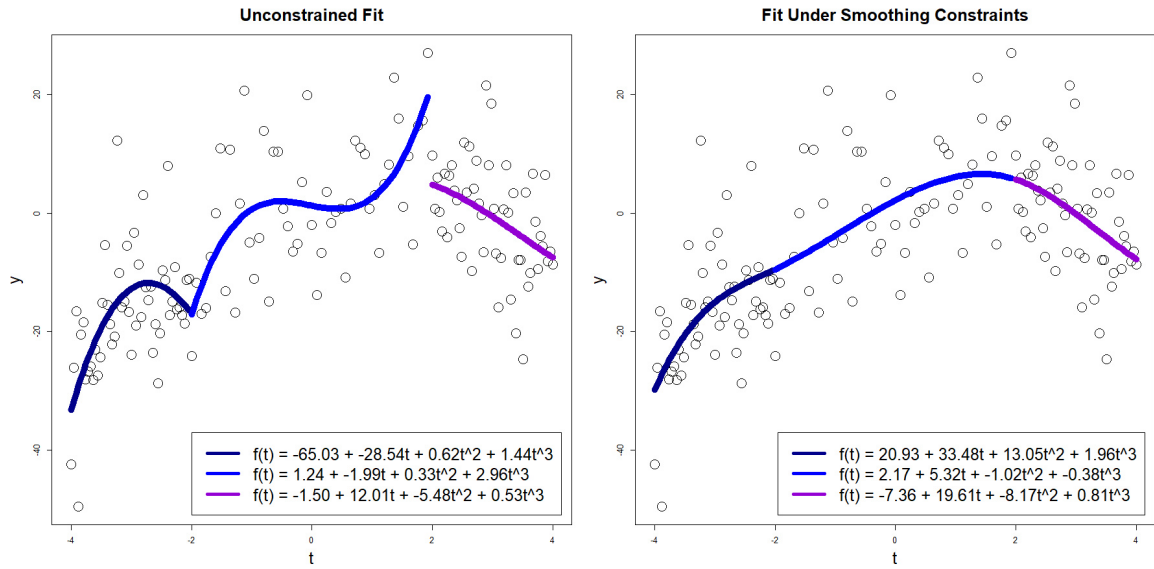


Figure 2: Example monomial cubic spline fit to data generated from a function of a single-predictor with independent, homoscedastic Gaussian distributed errors. *Left:* independent partition-wise fits, with discontinuities in fitted values and derivatives at the knots $t = \pm 2$. *Right:* the same data fit under the four-step transformed-OLS projection enforcing second-derivative, first-derivative, and fitted value continuity at both knots; each colored segment remains a cubic polynomial, and the three segments join smoothly.